Inorganic Cume
January 19, 2007

1. What are the advantages of using synchrotron radiation for structure determinations in contrast to conventional in-house tube or rotating-anode diffractometers?

2. What element of symmetry can be present in the lattice of a small molecule crystal that cannot be present in a protein structure crystal?

3. What is the reciprocal lattice? Contrast the reciprocal lattices of a small molecule structure and a protein structure.

4. In the common least-squares application the method is used to refine the parameters of slope and intercept for the best straight line fitting the data of a plot of points on a graph. In the least squares of the structural experiment, what are the parameters being refined and what data are being fitted?

5. Briefly (in a sentence or two) explain the following terms (found in a structural report):
   a. anisotropic displacement parameters
   b. isotropic displacement parameters
   c. R factor

6. What is the phase problem of crystallography?

7. Briefly describe how two of the four of the following are used to start a structural model (only answer for two of the four)
   a. Patterson maps
   b. Direct method calculations
   c. Molecular replacement (bio)
   d. MAD techniques (bio)

8. Neutron crystal data are more difficult to obtain than X-ray data. In what cases do people go to the extra effort to obtain neutron diffraction data?

9. Crystal structures are obtained by use of crystal reflection data. These data have a likely uncertainty of 5% or so. How can the results of the experiment give parameters such as bond distances of 1.8672 ± 0.0002? (i.e. how can the results have much smaller uncertainties than the original experimental data?)